



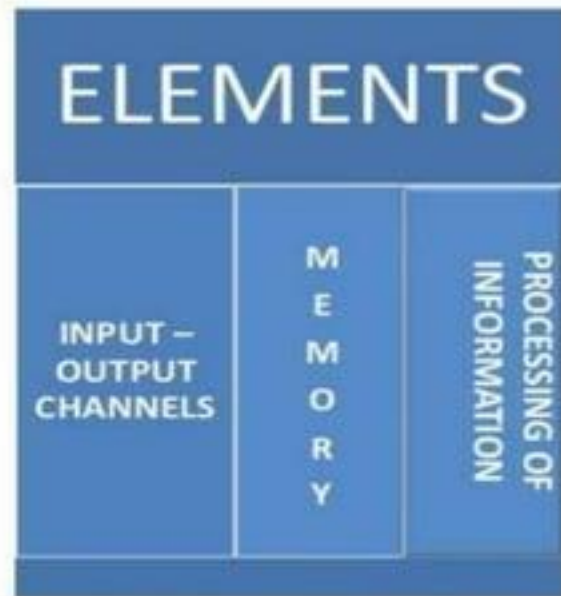
HUMAN COMPUTER INTERACTION

HUMAN MEMORY AND THINKING

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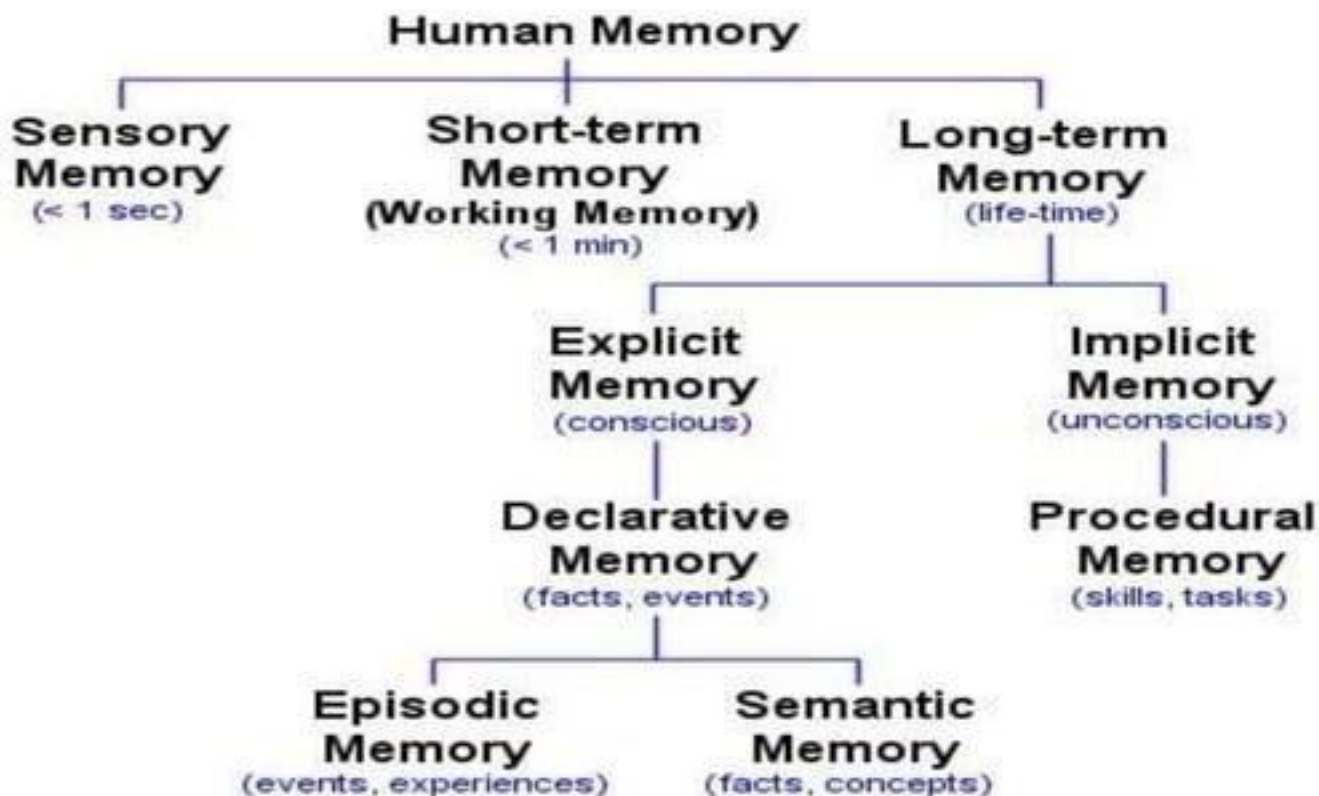
Overview

- Interaction with world
 - Occurs through information
- Interaction with computer
 - Input and output channels



- Then the information is stored in the memory
- Finally the information is processed and applied.
 - Reasoning
 - Problem solving
 - Skill acquisition
 - Error

Human Memory



Sensory Memory

- act as buffers for stimuli received through the senses
 - *iconic memory for visual stimuli,*
 - *echoic memory for aural stimuli*
 - *haptic memory for touch.*
- These memories are constantly overwritten by new information coming in on these channels.
- Examples – iconic memory
 - Moving a finger in front of the eye
 - “sparkler”
 - *Information remains in iconic memory very briefly, in the order of 0.5 seconds.*

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- Examples – *echoic memory*
 - direction from which a sound originates
 - Information is passed from sensory memory into short-term memory by attention, thereby filtering the stimuli to only those which are of interest at a given time.
 - Information received by sensory memories is quickly passed into a more permanent memory store, or overwritten and lost.

Short-term memory

- working memory
- acts as a 'scratch-pad' for temporary recall of information.
- used to store information which is only required fleetingly
- Example:
 - calculate the multiplication 35×6 in your head
 - 5×6 and followed by 30×6

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- Short-term memory can be accessed
 - rapid access $\sim 70\text{ms}$
 - rapid decay $\sim 200\text{ms}$
 - Two basic methods for measuring memory capacity.
 - determining the length of a sequence which can be remembered in order. **limited capacity 7 ± 2 chunks**
 - allows items to be freely recalled in any order.

Examples

212348278493202

0121 414 2626

HEC ATR ANU PTH ETR EET

chunking information can increase the
short-term memory capacity

Long Term Memory

- store factual information, experiential knowledge, procedural rules of behavior – Stores everything we know.
- Characteristics:
 1. It has huge capacity
 2. It has a relatively slow access time of approximately a tenth of seconds.
 3. Forgetting occurs more slow in long- term memory

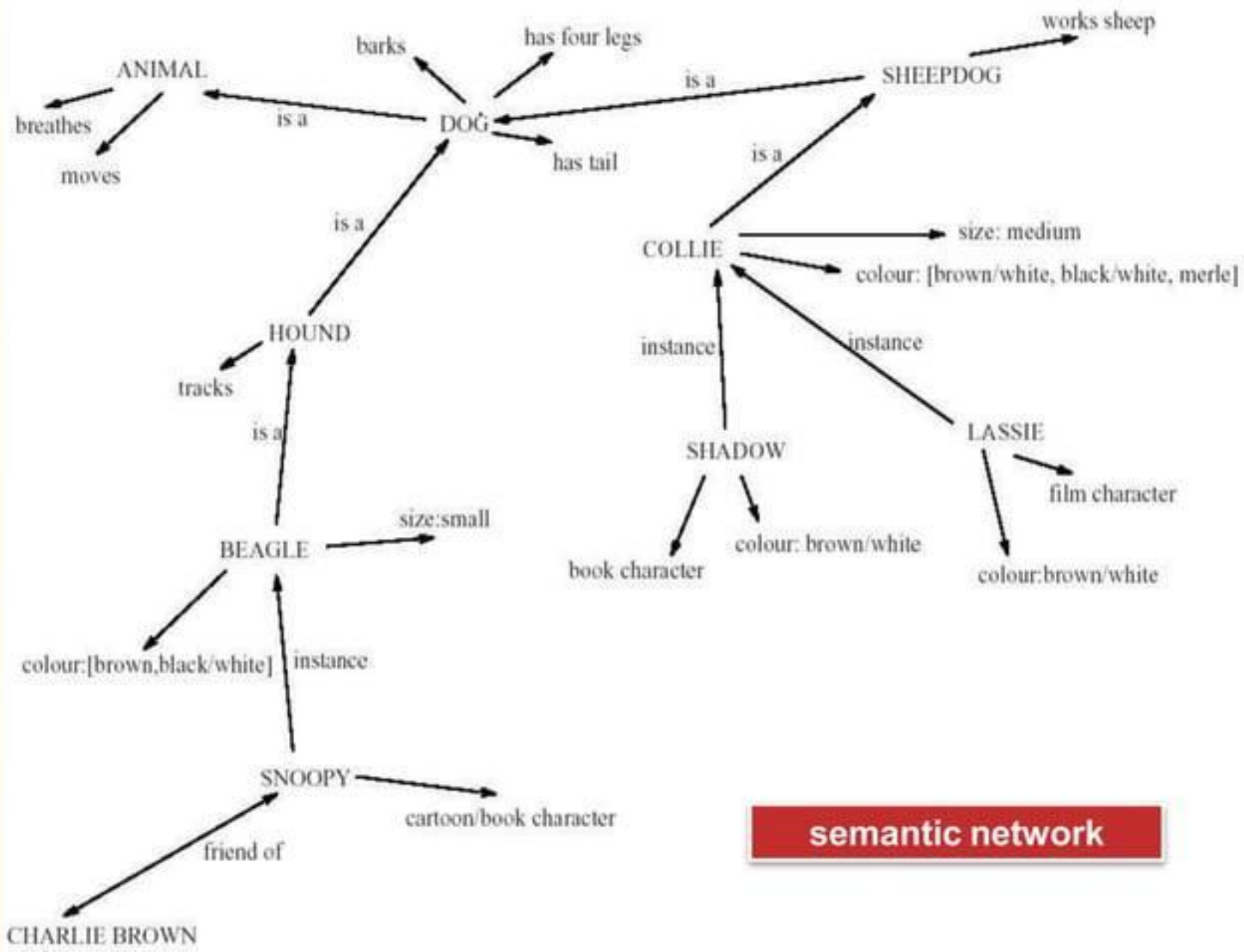
Long-term memory structure

- 2 types
 - *episodic memory*
 - represents memory of events and experience in a serial form.
 - Can recall an actual events took place at a given point of our lives.
 - *semantic memory*
 - structured record of facts, concepts and skills that we have acquired.
- semantic LTM derived from episodic LTM

LTM MODELS:

Semantic Network

- Semantic memory is structured as a network.
- Items are associated to each other in classes, and may inherit attributes from parent classes.
- Example:
knowledge about dogs may be stored in a network as shown

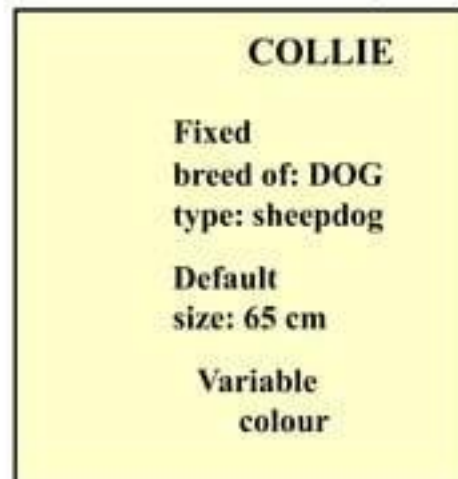


semantic network

LTM MODELS:

Frames

- Information organized in data structures
- Type-subtype relationships



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- Frame slots may contain default, fixed or variable information.
 - A frame is instantiated when the slots are filled with appropriate values.
 - Frames and scripts can be linked together in networks to represent hierarchical structured knowledge.

LTM MODELS: Scripts

- Scripts attempt to model the representation of stereotypical knowledge about situations.
- Ea: knowledge of the activities of doo

Script for a visit to the vet

Entry conditions:	<i>dog ill vet open owner has money</i>	Roles:	<i>vet examines diagnoses treats owner brings dog in pays takes dog out</i>
Result:	<i>dog better owner poorer vet richer</i>	Scenes:	<i>arriving at reception waiting in room examination paying</i>
Props:	<i>examination table medicine instruments</i>	Tracks:	<i>dog needs medicine dog needs operation</i>



A script comprises a number of elements, which, like slots, can be filled with appropriate information:

- **Entry conditions** Conditions that must be satisfied for the script to be activated.
- **Result** Conditions that will be true after the script is terminated.
- **Props** Objects involved in the events described in the script.
- **Roles** Actions performed by particular participants.
- **Scenes** The sequences of events that occur.
- **Tracks** A variation on the general pattern representing an alternative scenario.

LTM MODELS: Production rules

- Representation of procedural knowledge.
- Condition/action rules
 - if condition is matched
 - then use rule to determine action

IF dog is wagging tail THEN pat dog

IF dog is growling THEN run away



Long-term memory processes

- 3 main activities
 - Storage or remembering of information,
 - Forgetting
 - Information retrieval

Storage of information

rehearsal :

- Information is moved from short-term memory to long-term memory.
 - by repeated exposure to a stimulus or the rehearsal of a piece of information transfers it into long-term memory.
- repetition is not enough to learn information well. If information is not meaningful it is more difficult to remember.
- structure, meaning and familiarity
 - information easier to remember

Forgetting

- 2 main theories of forgetting:
 - *Decay*
 - *Interference.*

Decay

- information is lost gradually but very slowly

Interference

- new information replaces old: **retroactive interference**
- Ex: remembering your new phone number
- old may interfere with new: **proactive inhibition**
- Ex: find your self going to your old house instead of new one.

retrieval

- recall

- information reproduced from memory can be assisted by cues, e.g. categories, imagery

- recognition

- information gives knowledge that it has been seen before
- less complex than recall since the information is provided as cue



Thinking

Reasoning

deduction, induction, abduction

Problem solving

Reasoning

- Is the process by which we use the knowledge to draw conclusions or infer something new about the interest.
- inferring new information from what is already known
- Kinds of Reasons:
 - Deductive
 - Inductive
 - Abductive

Deductive Reasoning

- Deductive reasoning derives the logically necessary conclusion from the given premises.

e.g

- If it is Friday then she will go to work
It is Friday
Therefore she will go to work.

e.g. People from Pampanga cooks well and delicious
She is from Pampanga
Therefore she cooks well and delicious

Deductive Reasoning

- Logical conclusion not necessarily true:

e.g.

If it is raining then the ground
is dry It is raining
Therefore the ground is dry

Deduction (cont.)

- When truth and logical validity clash ...
e.g. Some people are babies Some babies
cry

Inference - Some people cry

Correct?

- People bring world knowledge to bear

Inductive Reasoning

- Induction:
generalize from cases seen to infer
information about cases unseen

e.g. all elephants we have seen have trunks therefore we
infer that all elephants have trunks.

- Unreliable:
 - can only prove false not true
- Humans not good at using negative evidence
e.g. Wason's cards.

Wason's cards



If a card has a vowel on one side it has an even number on the other

Is this true?

How many cards do you need to turn over to find out?

.... and which cards?

In fact, to test the truth of the statement we need to check negative evidence

Abductive reasoning

Abduction reasons from a fact to the action or state that caused it.

e.g.

Sam drives fast when drunk.

If I see Sam driving fast, assume drunk.

- **Unreliable:**

- can lead to false explanations

- If an event always follows an action, the user will infer that the event is caused by the action unless evidence to the contrary is made available.

- If, in fact, the event and the action are

Problem solving

- Process of finding solution to unfamiliar task using knowledge.
- There are a number of different views of how people solve problems.
- Several theories.

Gestalt Theory

- Problem solving is a matter of reproducing known responses or trial and error.
- problem solving both **productive** and **reproductive**
- Reproductive problem solving draws on previous experience.
 - Hindrance to finding a solution
- Productive problem solving involves insight and restructuring of the problem



Maier's *pendulum problem*





Problem space theory

- Proposed by Newell and Simon
- problem space comprises of problem states
- problem solving involves generating states using legal operators
- The problem has an initial state and a goal state and people use the operators to move from initial to the goal.
- heuristics may be employed to select operators

Sample Heuristic

means-ends analysis

- the initial state is compared with the goal state and an operator is chosen to reduce the difference between the two.
- Eg: reorganizing your office and you want to move your desk from the north wall of the room to the window
- Operators : carry or push or drag them
- If desk is heavy then new subgoal: to make it light.
- Operators : removing drawers and so on

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- An important feature of Newell theory is that it operates within the constraints of human processing system
 - so searching the problem space is limited by the capacity of short-term memory, and the speed at which information can be retrieved.
 - *General Problem Solver model* largely been applied to problem solving in well-defined domains, for example solving puzzles.
 - Also solving a programming problem- you need knowledge of the language and the domain in which the program operates

Analogy in problem solving

- how people solve novel problems.
- **analogical mapping:**
 - by mapping knowledge relating to a similar known domain to the new problem.



Procedure

- Notice the relationship
- Map source and target
- Apply mapping



Thank You